METRIC
MIL-DTL-24694A(SH)
w/AMENDMENT 1
18 January 2022
SUPERSEDING
MIL-DTL-24694A(SH)
4 October 2011

DETAIL SPECIFICATION

VALVE, HYDRAULIC RELIEF, 1.5 - 35 BARS (22 - 500 POUNDS PER SQUARE INCH) CARTRIDGE TYPE, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for cartridge-type relief valves for use in hydraulic systems with a maximum operating pressure of 35 bars (500 pounds per square inch [lb/in^2]).

1.2 Classification. Cartridge-type relief valves are furnished in the sizes, adjustment type, and pressure settings (classes) listed in the applicable specification sheet.

1.2.1 Part or identifying number (PIN). The PIN is constructed in accordance with the following:

M	**24694**	**/X**	**-**	**XX**	**-**	**XX**
Prefix for Military Specification	Specification Number	Spec Sheet Number		One- or two-character alpha to denote pressure range (see 3.1)		Two-character alpha to denote adjustment type (see 3.1)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip Wire, Forgings, and Structural and Special Shaped Sections

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-PRF-6083	-	Hydraulic Fluid, Petroleum Base, for Preservation and Operation
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-PRF-17672	-	Hydraulic Fluid, Petroleum, Inhibited
DELETED		
DELETED		
MIL-DTL-24694/l	-	Valve, Hydraulic Relief, 1.5 - 35 Bar, 150 LPM Cartridge Type
MIL-DTL-24694/2	-	Valve, Hydraulic Relief, 1.5 - 35 Bar, 10 LPM Cartridge Type
MIL-DTL-32353	-	Hydraulic and Lubricating Oil, Synthetic Hydrocarbon Base
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2193 - Ship Hydraulic System Components

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM20995 - Wire, Safety or Lock

NASM33540 - Safety Wiring, Safety Cabling, Cotter Pinning, General Practices for

(Copies of these documents are available online at www.aia-aerospace.org.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A582/A582M - Standard Specification for Free-Machining Stainless Steel Bars

ASTM B211/B211M - Standard Specification for Aluminum and Aluminum-Alloy Rolled or Cold Finished Bar, Rod, and Wire

ASTM B221M - Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE-AMS5610 - Steel, Corrosion and Heat-Resistant, Bars, Wire, and Forgings 12.5Cr - Low Carbon (SAE 51416, 51416Se) Free-Machining

SAE-AMS-QQ-S-763 - Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings

SAE-AS3209 - Packing, Preformed - AMS7276, 'O' Ring

SAE-AS4395 - Fitting End, Flared, Tube Connection, Design Standard

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Valves furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.3 Materials. Materials shall be limited to those specified herein and on applicable specification sheets. Metals shall be compatible with the fluid, temperature, service, and performance requirements specified herein.

3.3.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.2 Prohibited and restricted use materials.

3.3.2.1 Prohibited materials. The valves shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-020/077-2 or MIL-STD-2193.

a. DELETED.

b. DELETED.

c. DELETED.

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d. DELETED.

e. DELETED.

f. DELETED.

3.3.2.2 Restricted use material. Beryllium is acceptable as a small percentage alloying material in parts that are not subjected to welding.

3.3.3 Metal alloys. Unless otherwise approved by the qualifying activity, alloys for use in pressure-containing parts of valves shall be limited to the following:

a. Aluminum alloy: Alloy 6061 (UNS A96061) - Tempers T6, T651, or T6511 in accordance with ASTM B211/B211M or ASTM B221, as applicable.

b. Nickel-copper alloy: QQ-N-281, Class A (UNS N04400).

c. Corrosion resistant (CRES): Class 302 (UNS S30200), 304 (UNS S30400), 304L (UNS S30403), 316L (UNS 31603), or 440C (UNS S44004) in accordance with SAE-AMS-QQ-S-763, or Type 303 (UNS S30300) in accordance with ASTM A582/A582M, or Type 1 (UNS S41623) or 2 (UNS S41600) in accordance with SAE-AMS5610.

3.3.4 Fluid compatibility. The valve shall be compatible with the specified end-use hydraulic system fluid (see 6.2). Unless otherwise specified (see 6.2), the valve shall meet the requirements specified herein while operating with any hydraulic fluid in accordance with MIL-PRF-17331, MIL-PRF-17672, MIL-DTL-32353, or MIL-DTL-83282. Valves intended for use in systems with any of these hydrocarbon base fluids shall also be compatible with preservative fluid in accordance with MIL-PRF-6083.

3.3.4.1 Cadmium and zinc plating. DELETED.

3.4 Construction. Cartridge-type relief valve construction shall be as specified herein and on the applicable specification sheet. Overall dimensions shall be as specified on the applicable specification sheet.

3.4.1 Preformed packing (O-rings). O-rings shall be fluorocarbon rubber in accordance with SAE-AS3209.

3.4.2 Special tools. Valves shall be adjustable throughout the pressure range without the use of special tools. The adjustment type, either external or internal, shall be as identified in the applicable specification sheet (see 3.1). Special tools are defined as those tools not listed in the GSA Global Supply Catalog.

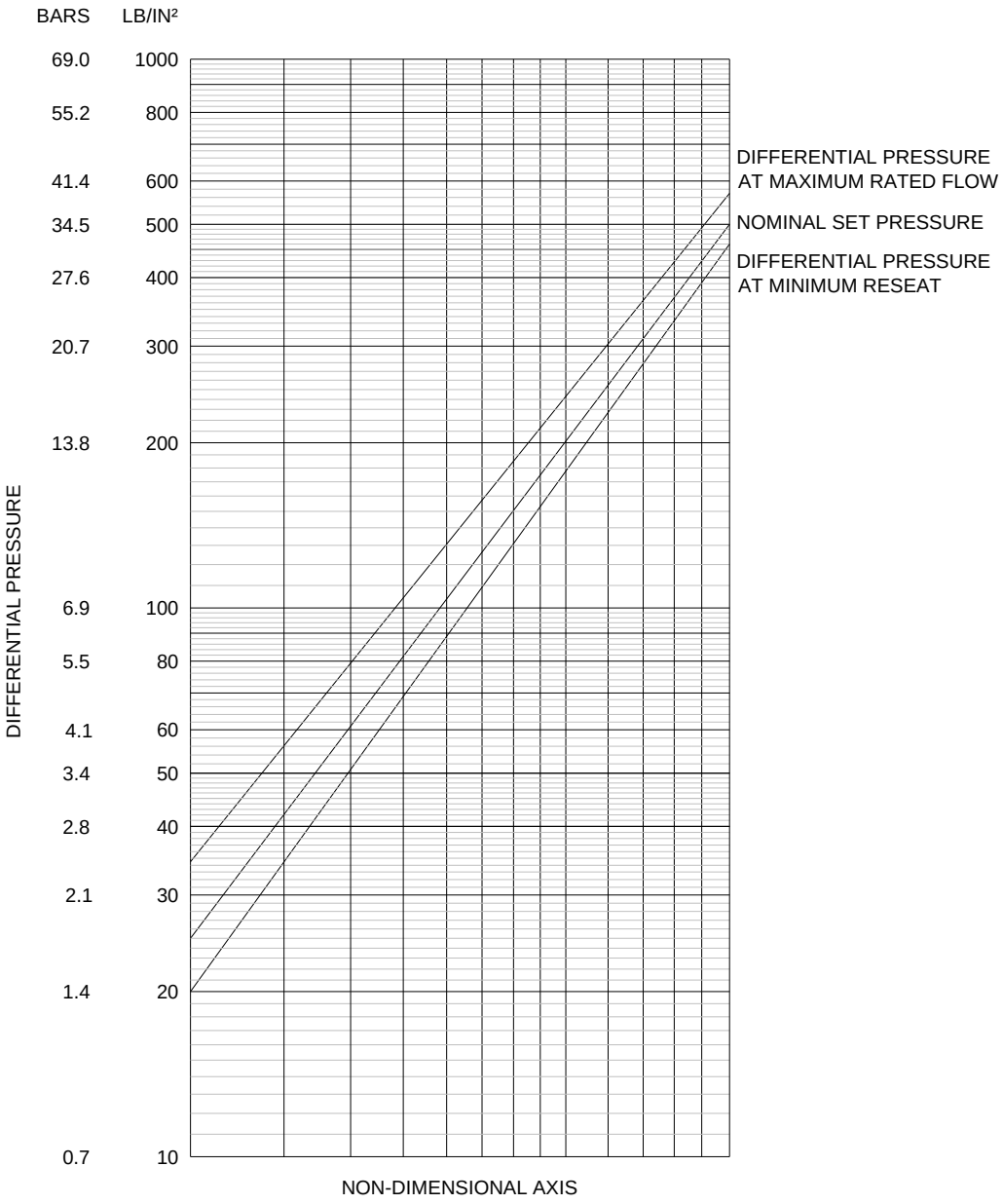
3.4.3 Safety. Hand-adjustable threaded parts subject to tampering shall be securely locked by safety wiring or self-locking nuts. Safety wire shall be applied in accordance with NASM33540 and shall conform to NASM20995. Star washers, lock washers, and jam nuts shall not be used as locking devices. Internal fasteners shall have features which prevent them from coming loose and interfering with system operation.

3.5 Performance. Valves shall perform as specified in 3.5.1 through 3.5.10.

3.5.1 Operating pressure. The valve shall meet the requirements specified herein at operating pressures specified in the applicable specification sheet (see 3.1) and specification part number (see 6.2).

3.5.2 Rated flow. Rated flow shall be in accordance with the values shown on the applicable specification sheet and shall occur at a differential pressure not greater than that shown on figure 1 for any nominal set pressure.

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NOTE:

1. Select differential pressure from the nominal set pressure curve, draw line vertically to obtain the differential pressure at maximum rated flow or at minimum reseat condition.

FIGURE 1. Low pressure hydraulic relief valve performance.

3.5.3 Proof pressure. The valve shall withstand a proof pressure of 69 bars (1000 lb/in^2) without external leakage, failure, or permanent set (see 4.6.2). If a spring is used that permits a pressure setting beyond 35 bars (500 lb/in^2), the proof pressure shall be 200 percent of the maximum pressure setting.

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3.5.4 Internal leakage.

3.5.4.1 Reseat. For any pressure settling throughout the applicable operating range, the valve shall reseat at not less than the minimum reseat pressure shown on figure 1. The valve is considered to be reseated when leakage does not exceed that specified in the applicable specification sheet (see 3.1).

3.5.4.2 Leakage. As pressure is decreased after reseating, the leakage shall not exceed 1 milliliter (0.061 cubic inch) per minute at a pressure equivalent to 75 percent of the set pressure. With pressure increased from zero to the minimum reseat pressure, the leakage shall not exceed 1 milliliter (0.061 cubic inch) per minute.

3.5.5 Endurance. The valve shall show no sign of damage to the seating surface, instability, or leakage in excess of twice the amount specified in the applicable specification sheet (see 4.6.7).

3.5.6 Vibration. The relief valves shall operate as specified herein after being subjected to exploratory, variable frequency, and resonance dwell sinusoidal vibration tests in each of three mutually perpendicular axes (see 4.6.5). Any of the following conditions resulting from vibration are considered unacceptable:

a. Damage to parts.

b. Loosening of parts.

c. Degradation of operational performance (including leakage) below specification requirements.

3.5.7 Shock. The valve shall meet the shock requirements as specified in MIL-DTL-901 for Grade A, Class I equipment (see 4.6.6).

3.5.8 Reseat pressure. The reseat pressure for any nominal set pressure shall be as shown on figure 1 (see 4.6.3.2).

3.5.9 Burst pressure. Unless otherwise specified (see 3.1), the valve shall withstand a burst pressure of 140 bars (2030 lb/in^2).

3.5.10 Operating temperature. Unless otherwise specified (see 3.1), the valves shall operate throughout a temperature range of -29 to 135 °C (-20 to 275 °F) when used with fluids suitable for the operating temperature.

3.6 Identification marking. Each valve shall be clearly and permanently identified by steel stamping, electrochemical etch, or equivalent means with the following information and symbol:

a. Manufacturer’s name, trademark, or logo.

b. Specification sheet part number.

c. Manufacturer’s part number.

d. Assembly date (by quarter and year).

3.7 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.8 Drawings. When specified (see 6.2), drawings shall display component assembly with parts identification/specification and provide pressure setting adjustment instructions.

3.9 Workmanship. The valve body and internal parts shall be free of burrs, sharp edges, or any other damage or defects that could prevent the part from performing its intended purpose. Valve shall be free of all metal chips, grit, dirt, and other foreign matter.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 Qualification inspection. Qualification inspection shall be performed on valves when qualification is required (see 3.2). Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEA. This inspection shall consist of the examinations and tests shown in table I. Tests shall be conducted in the sequence listed in table I unless otherwise approved by the qualifying activity.

TABLE I. Qualification inspection.

Examination or test	Requirement paragraph	Examination or text paragraph
Visual examination	3.3, 3.4, 3.6, 3.7, 3.8, 3.9	4.6.1
Proof pressure	3.5.3	4.6.2
Shock	3.5.7	4.6.6
Vibration	3.5.6	4.6.5
Performance testing	3.5	4.6.3, 4.6.4
Rated flow differential pressure	3.5.2	4.6.3.1
Reseat pressure	3.5.8	4.6.3.2
Internal leakage	3.5.4	4.6.3.3
Operating temperature performance	3.5.10	4.6.8.1, 4.6.8.2
Endurance	3.5.5	4.6.7
Burst pressure	3.5.9	4.6.9

4.2.1 Qualification sampling. Qualification inspection shall be conducted on one sample valve representative of pressure classes as specified on the applicable specification sheets.

4.2.2 Extension of qualification. Qualification may be extended on the basis of similarity if the following criteria are met:

a. Qualification of a valve corresponding to a specific pressure range may be extended to all ranges by similarity if only the springs are changed to alter the pressure range.

b. For the pressure range selected for qualification testing, the inspections shown in table I for qualification shall be accomplished.

c. For additional pressure ranges of the same configuration valve, only the rated flow differential pressure, reseat pressure, and leakage qualification tests shall be required.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.6.1 and the tests of 4.6.2 and 4.6.3.3. Each valve shall be subjected to conformance inspection.

4.4 Test conditions.

4.4.1 Test fluid. Unless otherwise specified herein, the hydraulic fluid used for all qualification tests shall conform to MIL-PRF-17672, 2135-T-H at a minimum temperature of 40 °C (104 °F). For conformance, any of the fluids specified in 3.3.4 may be used as the test fluid, provided that the fluid temperature is adjusted to maintain a fluid viscosity of less than 75 centistokes. Increased leakage for conformance testing with lower viscosity fluids shall be justified in the qualification test report and approved by the qualifying activity.

4.4.2 Ambient temperature. Unless otherwise specified herein, the tests shall be conducted at a room temperature of 15 to 35 °C (59 to 95 °F), measured within 0.3 meter (1 foot) of the test sample.

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4.5 Test methods. Cartridge-type relief valves shall be tested in a housing designed to accept the valve and having inlet and outlet ports. The valves shall be free of chatter, surging, or squeal under any condition or combination of conditions in this specification.

4.6 Examination and tests.

4.6.1 Visual and dimensional examination. Valves shall be visually and dimensionally examined to verify conformance to the requirements of this specification and those of the applicable specification sheet. The list of materials shall be inspected to verify that prohibited materials have not been used (see 3.3.2.1).

4.6.2 Proof pressure. A proof pressure of 69 bars (1000 lb/in^2) or 200 percent of the maximum possible pressure setting, whichever is greater, shall be applied to the inlet port of the valve with the return port blocked and held for 2 minutes. There shall be no external leakage or evidence of permanent deformation. For conformance, proof pressure shall be applied for a minimum of 30 seconds to assure that no leakage is present.

4.6.3 Performance tests (minimum setting). The following tests shall be conducted with the valve adjusted to the minimum nominal set pressure specified for its class as identified in the applicable specification sheet. Valves adjustable to a range of more than one class need only be tested at the minimum setting of the range. Any evidence of chatter, surging, or squeal under any test condition shall be cause for rejection. Leakage measurements shall be taken in the third minute of a 3-minute period. For conformance, test time may be reduced to 1 minute.

4.6.3.1 Rated flow differential pressure. The pressure shall be increased until rated flow as identified in the applicable specification sheet is passed through the valve. The pressure at which this flow occurs shall be recorded as the rated flow pressure and its value shall be not greater than that identified on figure 1 for a given nominal set pressure. Fluid temperature shall not exceed 40 °C (104 °F) for this test.

4.6.3.2 Reseat. With rated flow through the valve, the inlet pressure shall gradually be reduced until leakage not greater than the reseat leakage specified in the applicable specification sheet can be determined. The pressure at which this leakage rate occurs shall be recorded as the reseat pressure and its value shall be not less than that identified on figure 1.

4.6.3.3 Internal leakage.

4.6.3.3.1 Decreasing pressure leakage. The reduction of pressure shall be continued, and the leakage rate observed at a pressure equivalent to 75 percent of the set pressure. Unless otherwise specified (see 3.1), leakage at this pressure shall not exceed 1 cubic centimeter per minute.

4.6.3.3.2 Increasing pressure leakage. The pressure shall be decreased to zero, then increased to the minimum reseat pressure. Unless otherwise specified (see 3.1), leakage at the pressure equivalent to reseat pressure shall not exceed 1 milliliter per minute.

4.6.4 Performance test (maximum setting). The valve shall be adjusted to the maximum pressure setting of its class (see 3.1) as identified in the applicable specification sheet. Valves adjustable to a range of more than one class need only be tested at the maximum setting of the highest class. With the valve set at the maximum pressure setting, the tests of 4.6.3.1 through 4.6.3.3.2 shall be performed. Any evidence of chatter, surging, or squeal under any test condition shall be cause for rejection. Leakage measurements shall be taken in the third minute of a 3-minute period. For conformance, test time may be reduced to 1 minute.

4.6.5 Vibration tests. The valve shall be mounted on a resonance-free fixture for testing. The vibration input accelerometer shall be mounted on the fixture close to the mounting point of the valve to the fixture. A second accelerometer shall be mounted directly on the valve for monitoring resonant frequencies. Valves need not be pressurized for the tests but shall be mounted in a test block. If a valve is symmetrical about a principal axis, vibration testing need only be accomplished along two orthogonal axes, one of which shall be the principal axis of symmetry.

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4.6.5.1 Exploratory for resonance frequencies. Resonant frequencies of the valve shall be determined by varying the frequency of applied vibration slowly throughout the range specified in table II. For frequencies from 4 to 33 hertz, the table vibratory single amplitude shall be 0.25±0.05 millimeter (0.01±0.002 inch). Above 33 hertz, the table vibratory single amplitude shall be 0.10±0.02 millimeter (0.004±0.0008 inch). The change in frequency shall be made in discrete frequency intervals of 1 up to 50 hertz and maintained at each frequency for about 15 seconds. From 50 to 2000 hertz, a logarithmic sweep of at least 10 minutes duration shall be made. As an alternative, a logarithmic sweep from 4 to 2000 hertz back to 4 hertz may be conducted. The total time of the ascending plus descending sweep shall be not less than 40 minutes.

4.6.5.2 Variable frequency. The valves shall be vibrated from 4 (or lowest obtainable frequency) to 2000 hertz at the amplitude specified in table II. From 4 to 50 hertz, the valves shall be vibrated in discrete frequency intervals of 1 hertz for 1 minute minimum at each frequency. From 50 to 2000 hertz, a logarithmic ascending and descending sweep of not less than 20 minutes total shall be conducted.

4.6.5.3 Resonance dwell. Test valves shall be vibrated along each axis at the most severe resonant frequencies determined in 4.6.5.1 at the amplitudes specified in table II. The dwell test time shall be divided equally between the significant resonant frequencies. If more than four significant resonant frequencies are found for any one axis, the four most severe resonant frequencies shall be chosen for the dwell test. Total test time shall be 2 hours for each axis. If no resonant frequencies are identified, the valve shall be tested at 50 hertz.

TABLE II. Vibratory displacement, single amplitude.

Frequency range (Hz)	**Table amplitude** [1/]	
	(mm - minimum)	**(inches - minimum)**
4 to 15	1.25	0.05
16 to 25	1.00	0.04
26 to 33	0.75	0.03
34 to 40	0.50	0.02
41 to 50	0.25	0.01
51 to 2000	0.25	0.01

NOTE:

[1/] For Government-conducted tests, the maximum amplitude shall not exceed the minimum required amplitude by more than 20 percent.

4.6.6 Shock. The valve shall be subjected to high-impact mechanical shock tests in accordance with the requirements for Grade A, Class I, lightweight, Type B, fixture 4A of MIL-DTL-901. A valve shall be considered to have failed the shock test if it does not meet the leakage test of 4.6.3.3, or shows any signs of external or internal damage when examined in accordance with 4.6.1. The valve shall be pressurized to a setting not less than 75 percent of its maximum set pressure during shock testing.

4.6.7 Endurance. The valve shall be subjected to 100,000 cycles while set at the highest pressure setting for the class or group of classes. Each cycle shall consist of imposing rated flow through the valve and then reducing the pressure to zero. Cycling shall be accomplished at a rate not to exceed 60 cycles per minute. Fluid temperature shall be maintained at not less than 40 °C (104 °F). Leakage at the reseat pressure determined in 4.6.3.2 shall be not greater than two times the allowable reseat leakage identified in the applicable specification sheet.

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4.6.8 Operating temperature.

4.6.8.1 Low temperature. The fluid for this test shall be in accordance with MIL-PRF-83282. The test setup shall include a handpump or power-driven pump that shall provide a flow of not less than 1 liter (¼ gallon) per minute to the relief valve. The test shall be conducted on two valves, one adjusted to the lowest setting of the lowest pressure range tested and one adjusted to the highest setting of the highest pressure range tested. The pump shall be operated at room temperature at not less than the minimum flow rate. The highest pressure developed at the relief valve inlet shall be noted. The test setup shall then be maintained at a temperature not warmer than -29 °C (-20 °F) for 2 hours with the valve unpressurized. The valve shall then be pressurized to 75 percent of the set pressure and the low temperature maintained for 2 additional hours. The pump shall then be operated at the same rate as at room temperature and the highest pressure noted. The pressure value obtained at -29 °C (-20 °F) shall not vary more than ±10 percent from those obtained during the room temperature test.

4.6.8.2 Rapid warm-up and high temperature. The low temperature shall then be allowed to warm rapidly to a temperature of 135 °C (275 °F). While the temperature is being raised and without waiting for the temperature to stabilize throughout the setup, the pump shall be operated at the same rate as previously operated at room temperature at temperature increments not to exceed 30 °C (86 °F). The pressure values obtained at these temperature increments and at 135 °C (275 °F) shall not vary more than ±10 percent from those obtained during the room temperature test. After maintaining the 135 °C (275 °F) temperature for at least 2 hours, the reseat test of 4.6.3.2 shall be conducted with MIL-PRF-83282 fluid at 135 °C (275 °F) with a flow of not less than 1 liter per minute. The reseat pressure shall be not less than 90 percent of the minimum reseat pressure required by figure 1.

4.6.9 Burst pressure. Fluid pressure shall be applied to the inlet port with the outlet port blocked until the burst pressure specified in 3.5.9 or the applicable specification sheet (see 3.1) is obtained. This pressure shall be applied for a minimum of 2 minutes. At the end of this period, the valve shall be examined and any sign of external or internal damage shall be cause for rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valves described in this specification are intended for use in hydraulic systems operating at a maximum set pressure of 35 bars (500 lb/in² differential) and requiring a range of reseat and maximum accumulation pressures as shown on figure 1. The valves have an operating temperature range of -29 to 135 °C (-20 to 275 °F) and are intended for use with section 3.3.4 hydraulic fluids. User of this valve should confirm if a preferred orientation is necessary to obtain optimum performance within the intended application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of the applicable specification sheet.
c. Applicable specification sheet part number.
d. End-use hydraulic system fluid (see 3.3.4).
e. If drawings are required (see 3.8).
f. Packaging requirements (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24694 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil/.

6.4 Subcontracted material and parts. The packaging requirements of the referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Qualification

Threaded

6.6 Amendment notations. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4810-2022-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:02Z
Check the source to verify that this is the current version before use.